25 · Taste: the taste buds

The words of this track, as it says them. Bold brown is the examiner.

Taste. Five marks.

Briefly describe the structure of the taste buds, and discuss the physiology of taste sensation.

This one is her learning outcome, word for word. Her quizzes never asked it, which is exactly why it's worth having ready.

The hook: buds on the papillae, chemicals in the saliva, nerves seven, nine and ten, and up to the insula.

Taste, or gustation, is a chemical sense. Taste buds are chemoreceptors, stimulated by chemicals dissolved in the saliva.

The structure. The tongue carries papillae. Fungiform, which are mushroom-shaped. Vallate, surrounded by a moat. Filiform, thread-shaped. And foliate, leaf-shaped. The taste buds are specialised structures on the papillae. Each taste bud is a cluster of gustatory cells, about fifty to a hundred and twenty-five chemoreceptors, inside a capsule of supporting cells. Their gustatory hairs poke out into the taste pore.

Five tastes: sweet, sour, salty, bitter, and umami, which is savoury.

The pathway. The facial nerve, cranial nerve seven, carries taste from the front two-thirds of the tongue. The glossopharyngeal nerve, nine, carries the back third. And the vagus nerve, ten, carries the very back. They go to the brain stem, then up to the thalamus, the insula, and the frontal lobe. The amygdala and the hypothalamus add the emotional response, and appetite.

Flavour is taste plus smell. Chewing releases aroma molecules, which travel from the back of the mouth up into the nasal cavity, and the brain combines the two.

The traps. Taste buds are chemoreceptors. One of her true or false statements says they're photoreceptors. False. Second: filiform papillae carry no taste buds. They're for texture. The buds are on the fungiform, vallate and foliate papillae. Third: the facial nerve is the front two-thirds, and the glossopharyngeal is the back third.

Once more. Taste.

Taste is a chemical sense: taste buds are chemoreceptors for chemicals dissolved in saliva. They sit on the papillae, fungiform, vallate and foliate, and each bud is a cluster of gustatory cells in a supporting capsule, with gustatory hairs in the taste pore. Five tastes: sweet, sour, salty, bitter and umami. The facial nerve carries the front two-thirds, the glossopharyngeal the back third, and the vagus the very back, to the brain stem, the thalamus, the insula and the frontal lobe. Flavour is taste plus smell.